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A Web-Enabled Tool for Site Suitability Mapping for Managed Aquifer Recharge (MAR) Using Google Earth Engine (GEE) and Multi-Criteria Decision Analysis (MCDA)

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Abstract
Under possible risk of compound impacts of climate and anthropogenic changes, the Managed Aquifer Recharge (MAR) is envisaged to be an appropriate solution. Site suitability mapping is a crucial step in MAR project planning which requires various data, processing and GIS analysis. An automated web-based tool, named G-MCDA, is developed in this study for quick and efficient suitability mapping. G-MCDA integrates Google Earth Engine (GEE), Multi-criteria Decision Analysis (MCDA), GIS and web-based interface that allows automated data downloading, GIS data processing and suitability analysis using MCDA. G-MCDA has been tested in upper Ganga-Yamuna Doab region, India, for site suitability analysis. The application of G-MCDA shows that most of the area (82%) is under *suitable* category followed by *very suitable* (12%), and *moderately suitable* (5%). The suitable area is mostly flat agricultural lands with alluvium aquifer. A few patches of urban and sloping lands found to be *moderately/low suitable* for MAR.

Keywords Managed aquifer recharge · MAR · Groundwater recharge · Ganga-Yamuna Doab · G-MCDA

1 Introduction

Global groundwater use has increased manifold during past decades. In the future, water demand is expected to rise substantially which is expected to be intensified further by climate change. It is understood that the human interventions will affect groundwater more

Highlights
- A web-based tool (G-MCDA) is developed for site suitability mapping for Managed Aquifer Recharge (MAR).
- G-MCDA integrates Google Earth Engine (GEE), Multi-criteria Decision Analysis (MCDA) and GIS.
- Automated data retrieval module of G-MCDA allows on-cloud processing of 10 types of data.
- Application of G-MCDA in Ganga-Yamuna Doab, India, shows that 12% area is *very suitable* and 82% area is *suitable* for MAR.

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prominently than climate change (Li et al. 2020; Patidar and Keshari 2020; Scanlon et al. 2023; Dietmaier and Baumann 2023). Since groundwater is more resilient to short-term alterations in climate, the Managed Aquifer Recharge (MAR) is considered one of the adaptation measures. A variety of MAR techniques exists which can broadly be categorized as spreading methods, in-channel modifications, well injection, rainwater harvesting and induced bank filtration (Sallwey et al. 2019a; Alam et al. 2021). The percolation tanks, injection wells, infiltration ponds are some of the very common MAR techniques (Zhang et al. 2020). Whereas, the River Bank Filtration (RBF), Aquifer Storage and Recovery (ASR) and Sand Dams are amongst the growing techniques (Abbasi-Moghadam et al. 2021; Krishan et al. 2022). An effective implementation of MAR requires appropriate site selection, proper design of the structure and optimum operation. Several guidelines for MAR planning exist which elaborate site selection, design and operations (Dillon et al. 2014; CGWB 2020). Since the recharge rate is largely governed by physiography and hydrogeology which is often heterogenous in space, the identification of location where MAR structure should be built is crucial (Riyanto et al. 2020; Imig et al. 2022; Szabó et al. 2023).

Given that the site suitability analysis involves analysis of numerous data, Multi-criteria Decision Analysis (MCDA) with GIS is employed to quantify suitability levels for effective decision-making. Assigning weights to these criteria hinges upon their relative impact on recharge. For effectively estimate the weights for the criteria, a coherent framework is needed, leveraging criteria rankings, ratings, or relative importance. A diverse array of weight assignment techniques has been developed; Sallwey, et al. (2019b) provide an extensive overview of such methods.

While site suitability mapping can be conducted within GIS software, automated tools for quick suitability analysis are still not available Sallwey, et al. (2019a). Traditional approaches necessitate various data manipulations across GIS and other software platforms, specifically to convert temporal, spatial, and non-spatial data into thematic layers (Alam et al. 2020; Waterhouse et al. 2020; Sloan et al. 2023). Furthermore, the determination of weights is a distinct task. This approach involves a sequence of data processing steps distributed across multiple tools and software, often leading to prolonged timelines (Zhang et al. 2020). Despite its ability to yield the desired suitability maps, this desktop-oriented MCDA approach remains time-intensive.

Recent times have witnessed a notable surge in the adoption of web-based applications due to their user-friendly interfaces and easy accessibility. A noteworthy contribution by Sallwey et al. (2019b) appeared in the form of a web-based tool for site suitability mapping through MCDA. The advent of high-performance computing systems and cloud-computing infrastructures in the last decade has greatly enhanced the capabilities of web applications (Evans et al. 2020; Grasso et al. 2021; White et al. 2021). Simultaneously, abundant volume of data originating from remote sensing and models are now shared openly by various institutions and researchers, offering a valuable resource for site suitability analyses (Gorelick et al. 2017; Yang et al. 2022). Notably, Google Earth Engine (GEE), an ingenious cloud-based platform by Google, emerges as a potent platform that stores diverse spatial data and facilitates on-cloud data processing by using Google's high-performance computing infrastructure (Mutanga and Kumar 2019; Vos et al. 2019; Yang et al. 2022; Jaiswal et al. 2023). However, despite these advancements, integrated systems that encompass these advanced tools are not available for site suitability mapping for MAR.

The novelty of this study lies in the development of a web-based system that integrates GEE, GIS and MCDA for site suitability mapping for MAR. The system allows automated data retrieval and pre-processing on GEE, and establish a systematic data processing

approach for estimation of weights, standardization of raster layers and overlay analysis for MCDA. The developed system is named as "G-MCDA" which is openly available on the internet. In addition, a case study of Ganga-Yamuna Doab, India, is presented in this paper for site suitability mapping using the developed tool which provides valuable data for policy making.

2 G-MCDA Architecture and Interface

G-MCDA seamlessly integrates MCDA, GEE, and GIS within a Python environment. A dedicated module processes and retrieves GEE data on a server, utilizing GEE's Python API and built-in algorithms. It handles 10 types of GEE data using Python scripts developed in this study. For instance, the daily precipitation data (e.g., CHIRPS 5 km) is first processed to get the number of rainy days in each year and then the annual rainy days are aggregated over a period (say 30 years) to get a single band image. The final result is then rescaled to the required resolution and downloaded on the server for MCDA. Projected climate data can also be processed using a similar approach. This automated workflow eliminates local data handling and saves time and effort. GIS tasks are performed using Python libraries like *GDAL* and *Rasterio*. In the G-MCDA, reclassification, masking, and other image processing are automated, reducing the need for standalone GIS software.

The G-MCDA application for site suitability mapping involves weight estimation, standardization, constraint mapping, and overlay analysis. Three methods—rating, ranking, and Analytic Hierarchy Process (AHP)—are developed to estimate weights for MCDA. The processes—Client-side, GEE-side, and server-side—are linked by *Python*, *JavaScript*, and *C Sharp* scripts. Request from Client-side sends data for MCDA and inputs for GEE data processing (Fig. 1). Later, the processed GEE data is downloaded on the local server and the estimation of weight, standardization, and overlay analysis are performed.

The web interface encompasses seven important stages for site suitability mapping as depicted in Fig. 1. The initial sign-in/sign-up page facilitates user authentication and registration. The interface guides the user through various steps of MCDA with various options to get GEE data, standardize and assign weights. Upon completion of required steps, pressing the next button triggers the weighted overlay analysis and lead to an interactive map interface where the final suitability map and standardized maps are visualized. All data layers, including inputs, intermediate outputs and final outputs, are downloadable.

3 Material and Methods

3.1 Criteria Data

Number and type of criteria depend upon method of MAR, objectives of project and data availability. Sallwey et al. (2019b) categorizes MCDA criteria into four groups, i.e., aquifer, surface, water quality, hydrometeorology and management. The essential criteria may be identified based on their importance and data availability. Various thematic layers are required for site selection. Generally, one thematic layer corresponding to each criterion is used in GIS readable format, such as GEOTIFF. The G-MCDA tool helps creating such layers wherein data pre-processing is automated. There are a good remote sensing data archive on GEE which can be used for MCDA for MAR site selection. Additionally, the

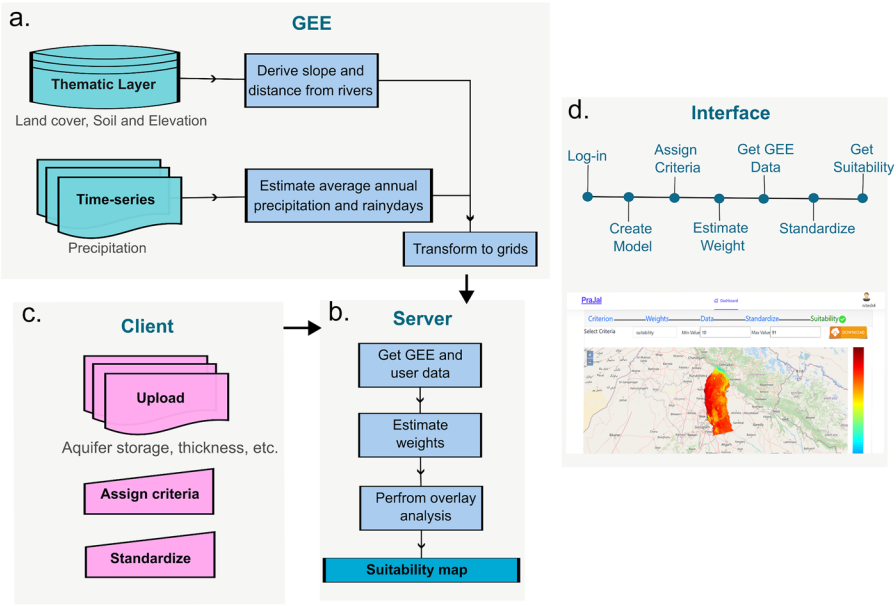


Fig. 1 Components of G-MCDA, major steps and interface layouts

users are allowed to upload additional data for desired outputs. The following sub-sections elaborate such data sets in detail.

3.1.1 GEE Data

GEE provides various remote sensing data acquired by different satellites, such as Landsat, Sentinel, MODIS, GPM, etc. In additional to the raw satellite data, various products which are derived from remote sensing data and modelling by different researchers and agencies are also available, such as precipitation, climate projections, population, etc. G-MCDA supports the automated pre-processing and downloading of the data listed in Table 1.

3.1.2 User Data

In many cases, the GEE data may not be enough to produce site suitability map with desired level of accuracy. Hence, G-MCDA offers the flexibility to incorporate supplementary data, ensuring the inclusion of crucial factors. Generally, these user-specified offline data predominantly pertain to aquifer properties, source water quality, and groundwater depth, enhancing the comprehensiveness of the analysis.

3.2 Assignment of Weights

The weights in MCDA indicate the relative significance of criteria governing the overall suitability of a site. Weight assignment is a crucial aspect of site selection; however, it is also perceived as complex and time-consuming. Based on applicability of wight estimation

Table 1 GEE data encompassed in G-MCDA for site suitability mapping

Sr. No.	Data ID	Description	Link to get more information
1	Land Cover Copernicus 100 m	Land cover map of 100 m resolution (CGLS-LC100)	https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_Landcover_100m_Proba-V-C3_Global
2	Precipitation CHIRPS 5 km	Annual precipitation at 5 km resolution derived from daily precipitation data of Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) from 2011 to 2020	https://developers.google.com/earth-engine/datasets/catalog/UCSB-CHG_CHIRPS_DAILY
3	Rainydays CHIRPS 5 km	Average number of rainy days at 5 km resolution counted from daily precipitation data of CHIRPS from 2011 to 2020	https://developers.google.com/earth-engine/datasets/catalog/UCSB-CHG_CHIRPS_DAILY
4	Slope srtm 90 m	Slope derived from SRTM 90 m data set	https://developers.google.com/earth-engine/datasets/catalog/CGIAR_SRTM90_V4
5	Population GPWv411	Gridded Population Version 4 (GPWv4), Revision 11	https://developers.google.com/earth-engine/datasets/catalog/CIESIN_GPWv411_GPW_National_Identifier_Grid
6	TWS GRACE Monthly Mass Grids—Land	GRACE-based monthly anomaly of TWS for the year 2016	https://developers.google.com/earth-engine/datasets/catalog/NASA_GRACE_MASS_GRIDS_LAND
7	Soil texture USDA	Soil texture data (USDA system) developed by EnvirometriX Ltd	https://developers.google.com/earth-engine/datasets/catalog/OpenLandMap_SOL_SOL_TEXTURE-CLASS_USDA-TT_M_v02
8	Significant Sens slope of ppt—CMIP5 RCP4.5	Significant Sen's slope of precipitation during 2021 to 2040 derived from NASA Earth Exchange Global Daily Downscaled Climate Projections	https://developers.google.com/earth-engine/datasets/catalog/NASA_NEX-GDDP
9	Significant Sens slope of rainydays—CMIP5 RCP4.5	Significant Sen's slope of number of rainydays during 2021 to 2040 derived from NASA Earth Exchange Global Daily Downscaled Climate Projections	https://developers.google.com/earth-engine/datasets/catalog/NASA_NEX-GDDP
10	Distance from river—Hydroshed River-based	Distance from pixel to river cell derived from river network data of HydroSHEDs	https://developers.google.com/earth-engine/datasets/catalog/WWF_HydroSHEDS_v1_FreeFlowingRivers

methods and current trends, we have identified three weight assignment methods: ranking, rating, and the AHP. These methods have been integrated into G-MCDA, featuring a user-friendly interface for efficient weight estimation. The following sub-sections elaborate on these methods in detail.

3.2.1 Rating

Rating is a simple method in which rates are assigned to criteria as per their importance. The weights are estimated based on the ratings using the following equation.

$$w_i = \frac{r_i}{\sum_{i=1}^{n} r_i} \tag{1}$$

where, w_i is weight of criterion i, r_i is the ranking assigned to criterion i, and n is the number of criteria. Using the G-MCDA, rating of each criterion can be assigned by typing a numeric value against each criterion. Estimated weights are displayed in the pie chart.

3.2.2 Ranking

In ranking method, rank of criteria is assigned as per their importance. The criteria with higher importance are ranked higher. G-MCDA allows estimation of weight using three ranking-based methods, i.e., sum, reciprocal and exponent based methods. The weight using these methods are estimated using the following formulae.

$$W_i = \frac{n - r_i + 1}{\sum_{r=1}^{n} n - r_i + 1} \quad \textit{Sum Weight} \tag{2}$$

$$W_i = \frac{{}^{1}/_{r_i}}{\sum_{r=1}^{n} {}^{1}/_{r_i}} \quad \textit{Reciprocal Weight} \tag{3}$$

$$W_i = \frac{(n - r_i + 1)^z}{\sum_{r=1}^{n} (n - r_i + 1)^z} \quad \textit{Exponential Weight} \tag{4}$$

where, r_i is rank assigned to criterion i, n is the number of criteria and Z is the exponent. User can select any method of ranking (sum, reciprocal or exponential), and accordingly the weights are displayed in the pie chart.

3.2.3 Analytic Hierarchy Process (AHP)

The AHP is a mathematical technique developed by Saaty in the 1970s (Saaty 1988). It organizes feelings and judgments into a hierarchy goals, criteria, and alternatives (Ghasemi et al. 2019). AHP is based on determining the hierarchy of problems and evaluating elements at hierarchical levels until the weight factors of all elements are determined. The AHP estimates weights using pairwise comparison matrix ($A_{(n\times n)} = [C_{ij}]$) which represents intensity of the expert's preference between individual criterion that affect the selection of the suitable placement. The judgment matrix is given in Eq. (2) for n criteria, where C_{ij} is the relative importance of the C_i criterion over the criterion C_j, where the values at diagonal (C_{ii}) indicates importance of a criterion against itself and is always equal to one.

$$\boldsymbol{A} = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1(n-1)} & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2(n-1)} & C_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{n(n-1)} & C_{nn} \end{pmatrix} \tag{5}$$

where $C_{ii} = 1, C_{ij} = \frac{1}{C_{ji}}$ *and* $C_{ji} \neq 0$

The relative weights of criteria $C_1, C_2, \ldots, C_n$ can be calculated from matrix **A**, where a normalization process is to be introduced. A new matrix is constructed by dividing the elements of each column in matrix **A** by the sum of the elements of the same column. The row averages of the new normalized matrix are then used to compute the relative weight of each criterion. After a certain number of pairwise comparisons, where some inconsistency can arise, consistency check of the judgments is necessary. The AHP includes a parameter to control the consistency of the weights, so called the Consistency Ratio (CR). For an acceptable result from AHP, the CR must be equal to or lower than 0.10. The CR is a ratio of the Consistency Index (CI) to the Random Index (RI).

$$CR = \frac{CI}{RI} \tag{6}$$

The CI can be calculated using Eq. (7):

$$CI = \frac{\lambda_{max} - n}{n - 1} \tag{7}$$

where λ_{max} is the eigenvalue of the pairwise comparison matrix and n is the number of the criteria. The RI depends upon the number of criteria, standard RI values available from literature can be utilized.

4 Application

4.1 Study Area: Upper Ganga-Yamuna Doab

G-MCDA is tested in the upper Ganga-Yamuna Doab region. This region is a part of Indo-Gangetic plain, spanning the area between rivers Ganga and Yamuna, and extends from Teria region in the North to Delhi in the South (Fig. 2). The total geographical area of the study covers 20,386 km^2. The river Hindon meanders through the middle of the study area and ultimately joins river Yamuna near Delhi. The area is densely populated, encompasses portions of Uttar Pradesh, Haryana, Uttarakhand, and Delhi. The terrain is predominantly flat, dominated by agricultural fields. Notably, the National Capital Region (NCR) Delhi, a bustling urban center and commercial hub of India, partly resides within the study area.

4.2 Criteria and Data

In the present case study, eight criteria — soil, slope, aquifer, land use and land cover, groundwater utilization, depth to groundwater, rainy days and distance from source — are identified considering their role and importance in governing the recharge processes. The aquifer type, groundwater utilization and depth to groundwater are uploaded while

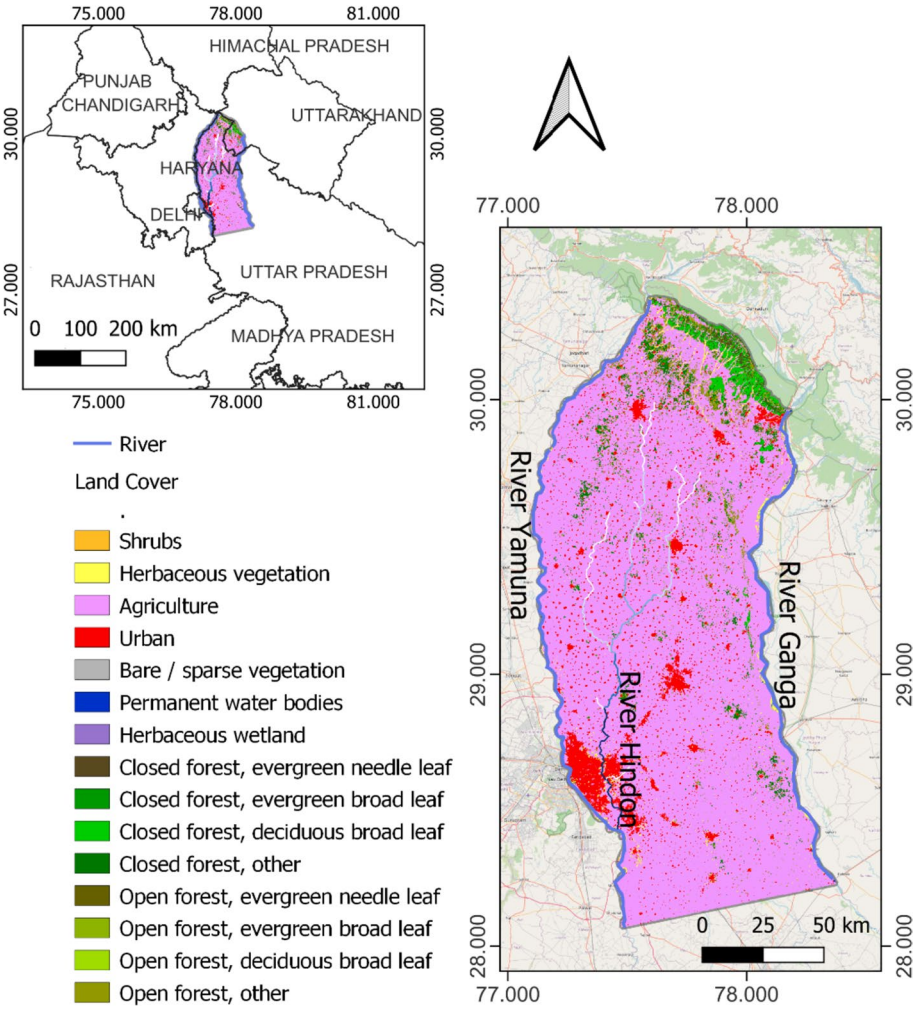


Fig. 2 Study area: Location and land cover

the other data sets were retrieved automatically from GEE. These data sets are depicted in Fig. 3.

4.3 Assignment of Weights

Weights can be determined using three distinct methods within G-MCDA: ranking, rating, or the AHP. To demonstrate capability of G-MCDA, all three methods, ranking, rating and AHP, are employed for estimation of weights. The ranks, ratings and pair-wise comparison matrix are given in Tables 2 and 3, and the resulted weights for each method is shown in Fig. 4. All these methods offer distinct logical framework for assigning importance of criteria, and therefore difference in estimated weights is obvious. Despite these variations, the relative importance of criteria remains consistent across all three methods.

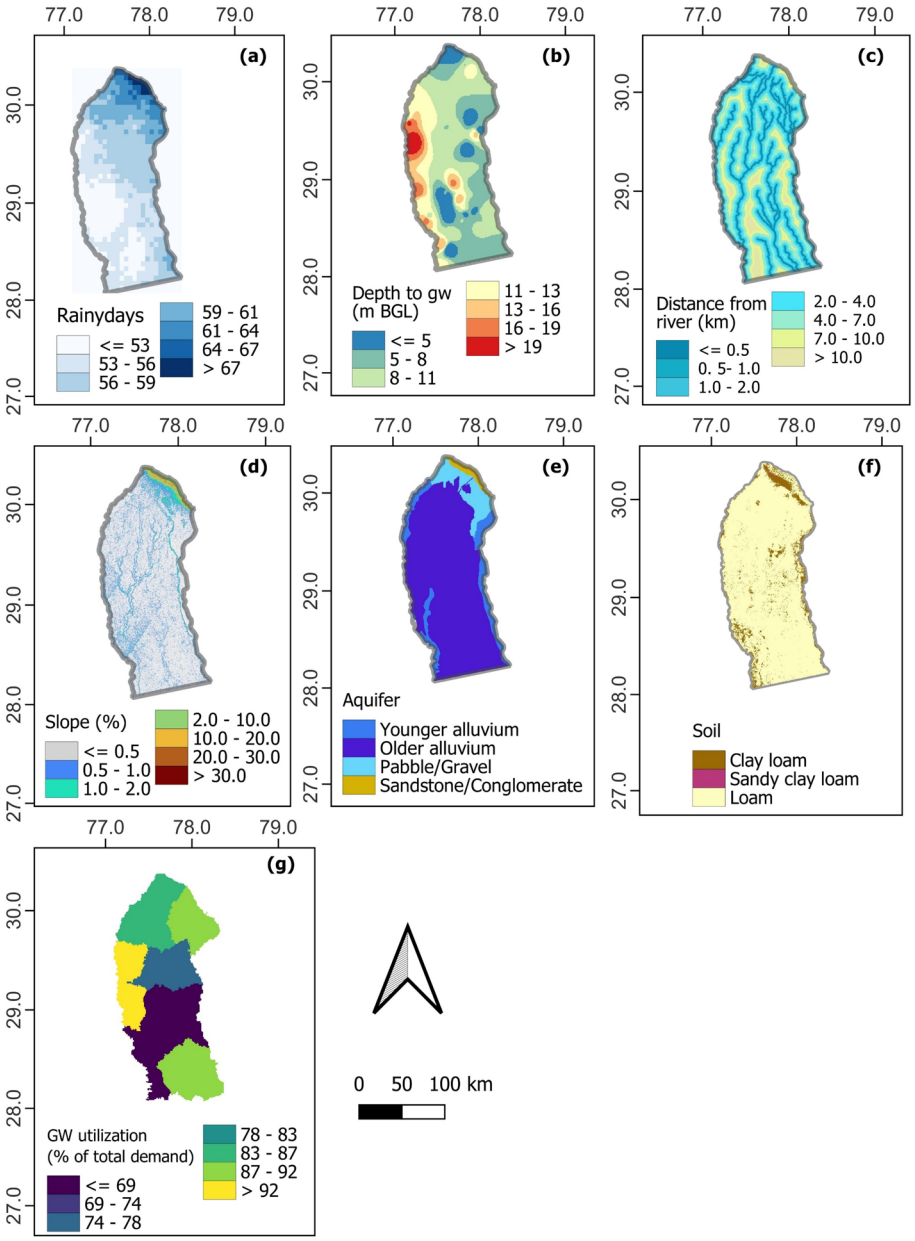


Fig. 3 Soil texture (**f**), groundwater utilization (**g**), depth to groundwater (**b**) and aquifer type (**e**), distance from river (**c**), Number of rainy days (**a**) and slope (**c**)

For example, the soil and slope get higher weights in all the methods. Weight for soil is estimated as 19% in ranking, 18% in rating and 18% in AHP. While ranking, rating, and AHP methods are compared for weight estimation, it is worth noting that the AHP id used for site suitability mapping.

Table 2 Ratings and rankings assigned to criteria

ID	Criteria	Rating	Ranking
1	Soil type	8	2
2	Slope	7	1
3	Aquifer type	5	4
4	LULC	8	3
5	GW utilization	5	7
6	Depth to GW	5	6
7	Rainy days	5	5
8	Distance from source	2	8

4.4 Standardization

Criteria layers are standardized to a value between 0 and 1 to perform overlay analysis for site suitability mapping. Linear transformation for standardization is prevalent approach in MCDA. It involves mapping original pixel values through a linear function to a standardized range, such as 0 to 1. For discrete layers, a direct translation to standard values is feasible. Each discrete class is assigned a standard value reflecting its significance, ranging from 1 (highest importance) to 0 (least importance). Notably, G-MCDA provides both linear transformation and discrete reclassification methods. The linear transformation option can also be utilized to create step-wise linear transformations. The standard values assigned to discrete layers are shown in Table 4. Meanwhile, Fig. 5 illustrates the linear functions utilized for standardization of continuous layers.

4.5 Site Suitability

The site suitability index signifies the degree of suitability, varying between 0 (*unsuitable*) and 100 (*very suitable*) (Fig. 6). This index can be further segmented into discrete categories, such as *very low suitable* (0 to 20), *low suitable* (21 to 40), *moderately suitable* (41 to 60), *suitable* (61 to 80), and *very suitable* (81 to 100). In the upper Ganga-Yamuna Doab, a substantial portion (82%)

Table 3 Pair-wise comparison matrix for AHP. The relative importance ranges from 1 to 9, wherein 9 indicates the highest importance

	Soil type	Slope	Aquifer type	LULC	GW utilization	Depth to GW	Rainy days	Distance from source
Soil type	**1**	1	1	1	2	3	3	4
Slope	1	**1**	2	2	2	3	1	4
Aquifer type	1	1/2	**1**	1	1	2	1	5
LULC	1	1/2	1	**1**	4	5	2	8
GW utilization	1/2	1/2	1	1/4	**1**	1	1	8
Depth to GW	1/3	1/3	1/2	1/5	1	**1**	1	9
Rainy days	1/3	1	1	1/2	1	1	**1**	9
Distance from source	1/4	1/4	1/5	1/8	1/8	1/9	1/9	**1**

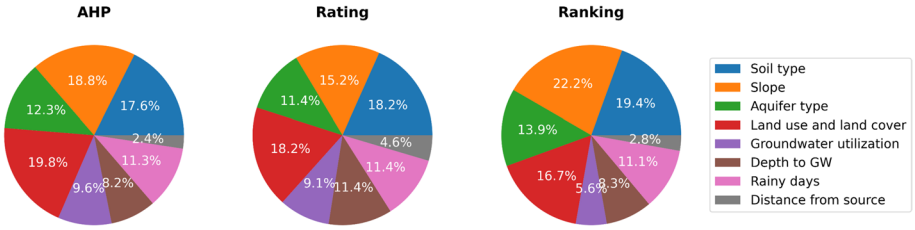


Fig. 4 Weights estimated using ranking, rating and AHP using G-MCDA

falls under the *suitable* category, followed by *very suitable* (12%), and *moderately suitable* (5%). Regions categorized as *very low suitable* and *low suitable* collectively account for less than 1%. The northern segment of the study area is designated as *low suitable* and *very low suitable* due to its high slope, dense forest cover, and unsuitable aquifer type. Similarly, urban zones are categorized as *low suitable*. The suitable regions primarily consist of flat agricultural land with an alluvium aquifer, rendering them potentially suitable for MAR projects for replenishing the Gangetic aquifer. Endowed with substantial aquifer storage capacity and water sourced from the Yamuna and Ganga rivers, this area emerges as a promising recharge zone. Nonetheless, considerations for water quality treatment are essential in MAR planning to avoid aquifer contamination.

Table 4 Standardized values of discrete layers (aquifer type, soil and land use land cover)

Aquifer type	Original pixel value	Standard value	**Soil type**	Original pixel value	Standard value
Younger alluvium	1	0.8	Clay loam	4	0.6
Older alluvium	2	0.7	Sandy clay loam	6	0.8
Pebble / Gravel	3	0.6	Loam	7	0.7
Sandstone / conglomerate	4	0.5			
Land use land cover			Original pixel value	Standard value	
Shrubs			20	0.75	
Herbaceous vegetation			30	0.75	
Agriculture			40	0.8	
Urban / built up			50	0.3	
Bare / sparse vegetation			60	0.9	
Permanent water bodies			80	0.1	
Herbaceous wetland			90	0.3	
Closed forest, evergreen needle leaf			111	0.65	
Closed forest, evergreen broad leaf			112	0.65	
Closed forest, deciduous broad leaf			114	0.65	
Closed forest, not matching any of the other definitions			116	0.65	
Open forest, evergreen needle leaf			121	0.75	
Open forest, evergreen broad leaf			122	0.75	
Open forest, deciduous broad leaf			124	0.75	
Open forest, not matching any of the other definitions			126	0.75	

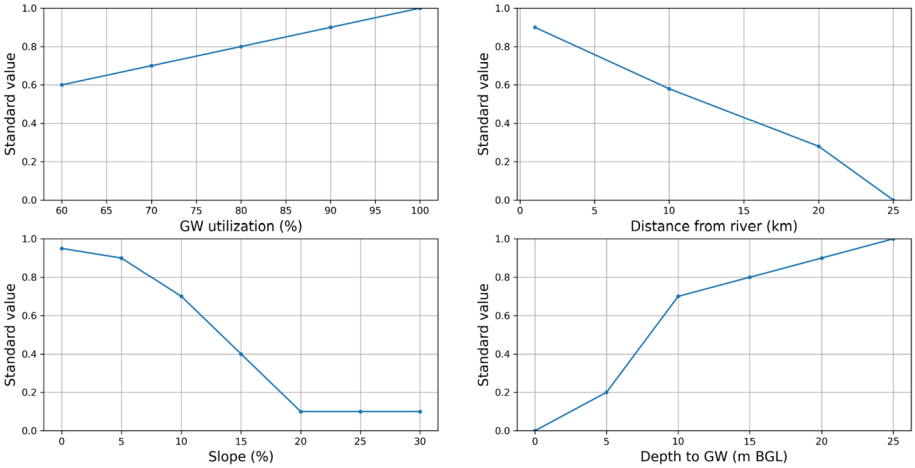


Fig. 5 Linear transformations for standardization of groundwater utilization, distance from river, slope and depth to groundwater

The western part, particularly near the Yamuna River, is classified as *very suitable*, making it ideal for MAR implementation aimed at stabilizing groundwater levels. By implementing MAR in this area, not only could groundwater decline be contained, but the availability of groundwater in Delhi NCR and the entire agricultural region in the central, southern, and eastern sections of the Ganga-Yamuna Doab could also be ensured. As groundwater flows in a southeastern direction within the region, recharge water would replenish not only the western parts but also bolster the aquifer in the southern and eastern regions.

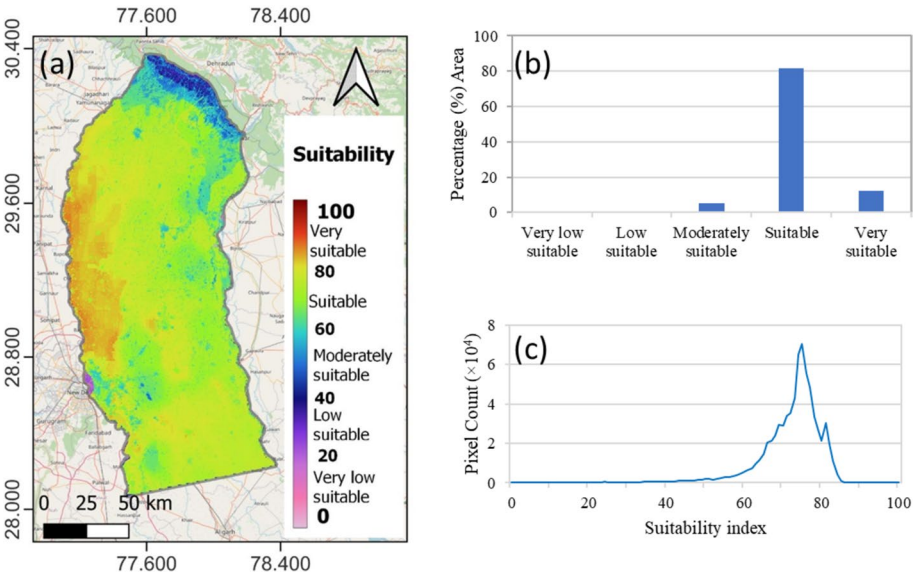


Fig. 6 Site suitability map (**a**) generated using G-MCDA, proportion of area under each category (**b**) and histogram of the suitability index (**c**)

5 Discussion

Identification of suitable sites using MCDA is governed by multiple factors, including the criteria, data and method of weight estimation. In addition, preferences of the criteria used in weight estimation influence the results considerably, and therefore participation of subject experts and decision makers is necessary to produce useful results from MCDA. However, site suitability mapping is often performed independently by modelers/GIS experts without the consideration of opinions from the subject experts and decision makers. One plausible explanation is that the integration of divergent expert opinions into MCDA necessitates multiple iterations of weight assignment and overlay analysis, a process that might not be efficiently executed through traditional methods.

Broadly, the sites which are hydrologically appropriate to increase recharge, suitable for constructing recharge structures and have sufficient good-quality water for recharge may be considered as suitable. Although, many factors influencing these aspects can be inferred from the spatial data, several are largely subjective. For instance, an agricultural landscape might be favored over an open forest by one expert, while another might prioritize the reverse. Addressing such conflicts is imperative to generate more meaningful MCDA outputs. Thus, an expedient site suitability mapping tool is essential to enable the iterative nature of MCDA using automated data retrieval and processing, and interactive visualization.

The G-MCDA tool, developed in this study, provides an efficient mechanism for reiterating and analyzing MCDA outputs, enhancing the efficacy of MAR planning. The present study demonstrates the capabilities of G-MCDA. The results are informed by the viewpoints of hydrologists and GIS analysts, signifying that including experts from diverse disciplines could mitigate biases in the results. Furthermore, the criteria assignment and their preferences are tailored to the specific problem.

6 Conclusions

The entire process of MAR planning involves subject experts, policy makers and stakeholders, and often needs reiterations of the processes to satisfy conflicting opinions of stakeholders and maximize the MAR outcomes. Site suitability mapping is typically done using desktop-based software wherein the data collection, processing and multi-criteria decision analysis are often time consuming. Automation of these steps could make the entire process of site suitability analysis very efficient. The developed G-MCDA tool automates data downloading and processing, and provides a quick way to estimate weights for MCDA using multiple methods. It is envisaged that with the use G-MCDA, site suitability mapping could be done more efficiently than other traditional approaches. The current version of G-MCDA includes 10 layers that can be downloaded automatically after partial processing at GEE's server and the additional data, such as aquifer thickness, specific yield, etc. may be uploaded by user. The application of G-MCDA in upper Ganga-Yamuna Doab, India, with GEE and some addition aquifer data, shows that the most of the area (82%) is under *suitable* category followed by *very suitable* (12%), and *moderately suitable* (5%). The suitable area is mostly under flat agricultural lands with alluvium aquifer. A few patches of urban and sloping lands found to be moderately/low suitable for MAR. The primary aim of MCDA implementation in this study was to showcase the web application's functionality, emphasizing that further refinement of results is imperative before full implementation. This entails incorporating input from decision-makers, as their preferences for specific criteria might influence the final outcomes.

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Author Contributions All authors contributed to the study conception and design. System development, data collection and analysis were performed by Dr. Nitesh Patidar and Dr. Basant Yadav. The back-end python scripts were developed by Dr. Nitesh Patidar. Mr. Sumit Kumar developed the front-end scripts for the system. All authors contributed in formulating the manuscripts.

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Software and Data Availability The G-MCDA was developed using Python, HTML and C# (C-Sharp). The back-end scripts for retrieving data from GEE, processing the thematic layers and performing overlay analysis using MCDA were written in Python. The Python scripts can be accessed through GitHub (https://github.com/niteshptdr/G-MCDA.git). The web-based interface of the tool is available on PraJal portal (http://prajal.org/tools.aspx) and can be accessed free-of-cost. The open-source data used in the study are available on GEE (https://earthengine.google.com/). **Software name:** G-MCDA. **Year first official release:** 2022. **Hardware requirements:** PC. **System requirements:** Windows, Linux or Mac with web-browser. **Availability: web-based tool**—http://prajal.org/tools.aspx. **Python scripts-** https://github.com/niteshptdr/G-MCDA.git. **Documentation:** manual available on http://prajal.org with example data sets.

Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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